
1 · Reference
Reference MD
(topology + coords)
Microstate atlas
grid · file · MSM
2 · Assign
CVs · assign i
π0(i)
π0(i)
3 · Symmetric perturbation (per frame)
ParmEd →
PyRosetta
Reference-frame
restraints
Pert: mutate
target
Ref: mutate /
enforce original
OpenMM (ff14SBonlysc + GBNeck2)
restrained min → U0, U1
ΔU = U1 − U0
4 · Reweight
fi = ⟨e^{−βΔU}⟩
ESS · QC · 𝒲
default
Optional:
neighbor pool · L-curve
π1(i), F1(i)